03_Using_Hog_svm_on_ATT_data_set

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Using HOD and SVM classifier:

Compute HOG features for each image.

Train an SVM classifier (e.g., from scikit-learn).

Test on unseen faces.

[1]:
```
from PIL import Image
import cv2
import numpy as np
from google.colab.patches import cv2_imshow
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

[ ]:
```
from skimage.feature import hog
from sklearn import svm
from sklearn.metrics import accuracy_score
import cv2, os, numpy as np

X, y = [], []
for i in range(1, 41):
    for j in range(1, 11):
        img = cv2.imread(f'/content/drive/MyDrive/
↪MCV_Continuatin_inGoogle_Colab/09_Face_detection/AT_T_database_of_face/s{i}/
↪{j}.pgm', cv2.IMREAD_GRAYSCALE)
        features, _ = hog(img, pixels_per_cell=(8,8), cells_per_block=(2,2),␣
↪visualize=True)
        X.append(features)
        y.append(i)
X, y = np.array(X), np.array(y)

clf = svm.SVC(kernel='linear', C=1)
clf.fit(X[:320], y[:320])
y_pred = clf.predict(X[320:])
print("Accuracy:", accuracy_score(y[320:], y_pred))
```

[ ]:
```
```